
BOAMP Observable-Successor Study

Complete Teaching and Defense Guide

From official notices to procurement episodes, successor linkage, survival analysis, trends, and technology classification

Scientific boundary. This study identifies observable successor procurements under a documented linkage rule. It does not certify legal contract renewals, causal effects, or individual forecasts.

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Technical summary

This project does **not** observe legal contract renewal. It reconstructs procurement processes from BOAMP notices, identifies later procurements that look like continuations of the same need, treats those algorithmically accepted links as **observable-successor events**, and studies their timing with survival analysis. A separate text classifier predicts business-technology classes for enrichment; it never defines a successor.

The canonical pipeline is internally consistent as of 22 August 2026. It standardises **1,620,712 unique notices**, reconstructs **1,103,632 episodes**, selects **3,800 awarded Grand Ouest episodes containing at least one CPV code in divisions 32/35/48/72**, exposes **763,417 anchor-candidate pairs**, and accepts **544 successors** under the frozen `M_B_text_ranking @ 0.70` rule. The resulting **14.32%** event rate and all Kaplan-Meier levels are linkage-conditioned, not renewal prevalence. The strongest comparative result is that CPV-35 has a higher observed-successor hazard than CPV-32 (HR **1.553**, 95% CI **1.218-1.981**), remaining **1.512** after candidate-pool adjustment. The Cox model is useful associationally but weak predictively: out-of-time C-index **0.479** for 2022-2024.

The technology model uses group-aware three-fold validation on **500 labelled notices in 459 procurement families**. Balanced multinomial logistic regression on word-TF-IDF achieves out-of-fold macro-F1 **0.744** (95% family-bootstrap CI **0.682-0.791**), versus **0.473** (**0.413-0.526**) for the best CPV/descriptor benchmark; paired improvement **0.271** (**0.201-0.340**). These are classification estimates on the annotation corpus, not proof that predicted class shares are market shares.

The evidence hierarchy is uneven. Structural pipeline integrity is strong. Comparative CPV-35 survival evidence is relatively robust. Absolute successor probabilities are sensitivity-dependent. Linkage accuracy is provisional because the reference labels are a single LLM research pass, subset spot-checked rather than independently specialist-reviewed. This project is descriptive and associational; **it does not establish causality or validated individual forecasting**.

1. The research problem before any model

Problem and why it matters

BOAMP is France's official bulletin for public-procurement notices. A row describes a publication: for example a call for competition, a correction, an award, or another procedural notice. It can carry a notice identifier, publication and deadline dates, buyer name and sometimes SIREN/SIRET, postal geography, CPV codes, object and description text, procedure and framework wording, declared durations, lots, and amount candidates. It does **not** provide a reliable field saying “this new procurement replaces that earlier contract.”

The real-world phenomenon of interest is recurrent demand: after a public buyer procures a digital need, when does a later procurement appear that plausibly continues or replaces it? The ideal dataset would contain a stable contract identifier, buyer and supplier legal identities, signed and expiry dates, option exercises, amendments, termination, and an explicit predecessor/successor or renewal relationship. BOAMP instead observes publications. Legal renewal can occur without a new BOAMP notice, and a new notice can be a re-tender, replacement, extension, new phase, parallel lot, or unrelated purchase.

Idea and why it makes sense

The project constructs an observable proxy:

An **observable successor procurement** is a later BOAMP procurement episode from the same plausible buyer that appears to continue or replace the same need and is accepted by a frozen linkage rule.

“Observable” says the event must leave a BOAMP trace. “Successor” avoids claiming a particular legal mechanism. “Procurement” says the matched object is a later buying process, not necessarily an amendment to the same legal contract.

The concepts are distinct:

Concept	Meaning	Directly observed here?
Legal renewal	Exercise or continuation under contract law	No
Procurement continuation	The same substantive need continues	Not directly
Successor procurement	A later buying process replaces/continues a need	Inferred
Observable successor in BOAMP	A later episode accepted by the documented linkage rule	Constructed

The legitimate claim is about **time to an accepted observable successor under a specified rule**. The study cannot claim that all 544 links are legal renewals, that censored episodes were never re-procured, or that hazard-ratio differences are causal.

The whole logic in one sentence is:

Because we cannot observe **legal renewal** directly, we construct **observable-successor procurements** using **standardised BOAMP episodes, buyer/chronology blocking, and frozen text-based linkage**, and then analyse **right-censored time to that constructed event**.

Quantity and goal labels

Observed quantities come from BOAMP or annotation: notice date, raw buyer identifier, CPV, text, reviewed label. Constructed quantities are deterministic or algorithmic: standardised field, episode, candidate set, accepted link, censoring row. Estimated quantities come from statistical procedures: Kaplan-Meier probability, hazard ratio, confidence interval, trend slope. Predicted quantities come from machine learning: technology class and confidence score. Keeping these labels separate prevents a prediction from being retold as an observation and a proxy from being retold as legal truth.

2. Complete project map and why every arrow exists

Official BOAMP JSONL, 2015-2025

↓ schemas differ; raw fields are not comparable

Schema-aware notice standardisation

↓ one procurement can publish several notices

Procurement episode reconstruction

↓ the research scope is awarded Grand Ouest digital demand

Awarded Grand Ouest any-digital-CPV cohort

↓ a successor must be searched for among later episodes

Buyer-and-time candidate generation

↓ blocking only creates plausible comparisons; it does not decide

Four linkage methods on the same pairs

↓ survival requires one event or an abstention per anchor

M_B top-text candidate accepted at 0.70, otherwise abstain

↓ episodes without an accepted link still contain follow-up information

Right-censored survival dataset

↓ censored observations invalidate ordinary event-only averages

Kaplan-Meier, Cox, parametric and sensitivity analyses

Parallel enrichment branch:

500 labelled procurement-object texts

↓ related notices can leak across ordinary random folds

459 procurement families and grouped cross-validation

↓ text must become numeric evidence

word TF-IDF + balanced multinomial logistic regression

↓ predictions are uncertain and some classes are weak or small

out-of-fold validation, bootstrap, calibration decision and quality gates

↓ only defensible predicted groups should be analysed

technology-specific survival and quarterly trend enrichment

The trend branch asks a different time question:

awarded episodes → award quarter → quarterly counts → OLS / PELT / ADF-KPSS / HMM

Survival time is age since an anchor award; trend time is calendar quarter. Mixing them would confuse “when a need reappears after award” with “how total procurement activity changes through history.”

3. Raw notices: why one publication is not one procurement

Problem

Suppose a city launches a two-lot IT tender in March, corrects the response deadline in April, publishes an award in July, and publishes separate lot information. Those four publications are administrative views of one underlying process, not four independent purchasing needs.

Practical example

Notice A, 10 March: call for tenders – finance software + training

Notice B, 2 April: correction – deadline changed

Notice C, 20 July: award result

Notice D, 21 July: award details for lot 2

At notice grain, B appears 23 days after A and C one hundred days later. A notice-level successor algorithm could link A→B or B→C and invent very short “renewal” times. It could also count lots separately and let an award notice link to its own consultation.

Output, interpretation, limitation, next step

Raw standardisation keeps one row per official notice. The project has **1,620,712** such unique rows from 2 March 2015 through 31 December 2025, including **209,502** geolocated to Grand Ouest. This is an observed publication layer, not yet a procurement-process layer. The next step must reconstruct episodes so repeated administrative observations do not become demand events.

4. Standardisation: making heterogeneous schemas comparable

Core idea

The raw BOAMP format changes across legacy XML-derived schemas, simplified forms and eForms. Standardisation uses field-specific parsers and keeps both the raw value and its lineage. “Unresolved” is preferred to a generic JSON guess because a wrong buyer, date or amount becomes strong false evidence downstream.

Field	Raw problem	Rule and mechanism	Example	Error prevented
Dates	strings, timestamps, invalid dates	parse an explicit YYYY-MM-DD pattern; normalise datetimes to UTC; flag impossible order	2025/04/03 → 2025-04-03	negative durations, wrong chronology
SIREN/SIRET	punctuation, wrong length, arbitrary IDs	keep digits; validate 9- or 14-digit identifiers with Luhn; derive SIREN from valid SIRET	123 456 789 00012 → validated SIRET + first 9 digits as SIREN	false legal-entity match
Buyer name	case, accents, punctuation, schema variants	Unicode normalisation, case folding, punctuation/space collapse; legal terms retained at standardisation	VILLE DE NANTES → ville de nantes	missed same-buyer comparison
CPV	main and lot codes live in different paths	find eight-digit CPV patterns across schema-aware locations; preserve all unique codes and identify a preferred main code	main 45000000, lot 72000000 → both retained	excluding a digital lot

Field	Raw problem	Rule and mechanism	Example	Error prevented
Text	object/title/description scattered	collect object, titles and descriptions; normalise whitespace; deduplicate	repeated title appears once	noisy or empty similarity input
Geography	buyer vs service-location fields differ	use contracting-buyer structured postcode, derive department and Grand Ouest region	44000 → department 44, Pays de la Loire	wrong regional cohort
Framework	inconsistent explicit fields	search normalised procurement text for <code>accord cadre / framework agreement</code>	phrase found → flag 1	missing a Cox covariate
Duration	days, months, years, start/end dates; conflicts	convert typed measures to months; require plausible 0-240; consensus or conflict; never global-impute	730 days → 23.98 months	fabricated expiry dates
Missing values	empty strings can resemble agreement	encode absence as empty/NaN; in weighted scoring, missing evidence drops out	both SIRENs missing ≠ match	“unknown = same” error
Duplicates	repeated file extraction	enforce unique official <code>idweb</code>	duplicate ID count = 0	double counting
Lineage	cleaning can hide source	retain <code>raw_source_file</code> , raw fields, parser version and record SHA-256	transformed value traceable to raw record	unauditable corrections

Validated identifiers dominate names because they identify legal units. If two validated SIRENs conflict, similar names must not override them: “Ville de X” and “Métropole de X” can share tokens while being legally distinct. The canonical checks find **zero accepted links with conflicting validated SIRENs** and **zero municipality/intercommunal mixes**. The limitation is that validated SIREN is missing for $2,521/3,800 = 66.34\%$ of cohort episodes, so name-based identity remains necessary and weaker.

Reliable duration is present for only $953/3,800 = 25.08\%$ and missing for $2,847/3,800 = 74.92\%$. Its completeness rises from 11.8% in 2023 to 84.4% in 2025, proving that missingness changes with schema/time. A universal four-year imputation would encode an assumption as data.

5. Episode reconstruction: the first constructed research object

Problem and idea

The award notice alone may omit original text, consultation timing, complete CPVs, corrected buyer details or lot structure. The idea is to treat notices as nodes in a graph and trustworthy relationships as edges. Every connected component becomes an inferred procurement episode.

Method

Edges are considered in decreasing reliability:

1. shared `contractFolderID`;
2. explicit BOAMP links to earlier notices;
3. exact normalised procedure reference for the same buyer within 730 days.

Before joining two components, a validated-buyer conflict vetoes the merge. A union-find algorithm joins accepted edges efficiently. The episode then aggregates all notice IDs, earliest substantive competition date (or earliest notice), latest publication date, natures, CPVs, deduplicated text, buyer evidence, region, framework flag and consistent duration.

Worked example

A consultation -explicit link- B correction -same folder- C award
+same reference/buyer- D lot result

All four become episode E1. A later 2023 consultation E2 remains a separate component because it has a new folder/reference and is years later. E2 may be a successor; B cannot be, because B is already inside E1.

Actual calculation and output

1,620,712 notices become **1,103,632 episodes**, a reduction of **517,080 publications (31.9%)** at the change of grain. Episode sizes are 662,798 singletons, 379,750 with two notices, 50,423 with three, 8,387 with four, and 2,274 with five or more. Accepted reconstruction evidence includes 45,885 folder edges, 584,896 explicit-link edges and 195,043 exact-reference/same-buyer edges. All notices are assigned once; there are zero buyer-conflict episodes and zero impossible chronologies.

This prevents duplicate events, self-linkage, artificially short gaps and administrative republication being counted as renewed demand. It does **not** prove semantic perfection: episode reconstruction is inferred, and 3,344 unusual/conflicted cases are exported for review.

6. Cohort construction and its selection effects

Filtering chain

Sequential filter	Episodes remaining	Removed at step
Grand Ouest reconstructed episodes	144,269	-
at least one CPV in 32/35/48/72	7,376	136,893
carrying an award notice	3,826	3,550
resolvable award date by cutoff	3,800	26

The unit is one awarded episode. Grand Ouest supplies a coherent operational territory; award status provides a defensible time origin; the CPV rule narrows to digital-related demand; a valid award date is required to measure elapsed time. Each restriction may select procurements with better reporting and excludes needs purchased or published differently.

For episode i , let CPV_i be all its codes. The indicator is:

$D_i = 1$ if there exists c in CPV_i whose two-digit division is 32, 35, 48 or 72;
 $D_i = 0$ otherwise.

“Any-code” differs from “main-code-only.” Example: a construction framework has main CPV 45000000, lot 1 electrical work, and lot 2 network integration CPV 72000000. Main-only exclusion calls the whole episode non-digital; any-code inclusion retains the digital lot. In the actual cohort, $1,176/3,800 = 30.95\%$ enter although their main CPV is outside the digital set. Therefore the precise name is “awarded Grand Ouest episodes containing at least one qualifying digital CPV,” not simply “3,800 digital contracts.”

For mutually exclusive segment curves, $digital_segment$ is the lowest qualifying division. It affects $412/3,800 = 10.84\%$ with multiple digital divisions. Among episodes with a digital main CPV, it agrees with the main division 94.7%; the measured sensitivity is small, but the tie-break is still arbitrary.

7. Successor notation and candidate blocking

Let i be the awarded anchor episode; j a later episode; u_i the anchor award date; v_j the candidate's first-publication date; and J_i the candidates surviving blocking. The problem is to choose one j in J_i , or choose none, that plausibly continues the same need.

This is record linkage/entity resolution because there is no labelled outcome column to predict directly; two records must be judged as representing a substantive predecessor/successor relationship. Abstention is essential because many anchors have no plausible observed successor.

Why blocking exists

Exhaustive comparison would create billions of obvious non-matches and many chances for an unrelated text to score highly. Blocking asks only “worth comparing?”, never “is the successor?” Candidates require:

- same buyer key or same normalised buyer name;
- no conflict between two validated SIRENs;
- candidate not equal to anchor;
- $u_i + 90 \text{ days} \leq v_j \leq u_i + 2,920 \text{ days}$.

Municipal prefixes such as “commune de” may be removed for blocking, but intercommunal legal forms are preserved. Geography is already constrained by the Grand Ouest pool and is not an acceptance feature. CPV is deliberately **not** a hard block: **9/23 reviewed successors cross CPV divisions**; hard same-division blocking would reduce the attainable ceiling from 0.913 to 0.609.

The 90-day lower bound removes parallel lots and concurrent programmes; the shortest reviewed successor gap is 139 days, so it removes none of the 23 reviewed positives. The 2,920-day upper bound bounds computation and implausibly remote matches and excludes none of those reviewed successors. It is not an eight-year contract assumption.

Actual output: **763,417 pairs for 3,520/3,800 anchors (92.63%)**; 280 anchors have no candidate. Among covered anchors, pool size ranges 1-1,713, median 89, mean 216.9. A large block raises both computation and the chance that a maximum similarity clears the threshold.

8. Candidate-generation recall and its ceiling

There are two gates at which a true successor can disappear:

```

true successor
  ↓ was it exposed by blocking?
candidate set
  ↓ did scoring rank and accept the correct row?
accepted successor

```

Blocking recall is:

```

known successors present in their candidate sets / known successors
= 21 / 23
= 0.9130 = 91.3%.

```

No scoring model can recover the two unreachable successors, so 91.3% is a reference-sample recall ceiling. One case never entered the cohort because the award lacked structured Grand Ouest geography; one buyer changed legal form from CCAS to CIAS without a shared SIREN. These are explained operational losses, not scoring bugs.

The limitation is important: the reference reviewer saw roughly 25 surfaced candidates per anchor, but the rule that produced that review list is not recorded. Labels are independent of the later linkage methods, but candidate-surfacing independence is not established. Therefore the 21/23 ceiling and recall estimates are less independent of text ranking than precision is.

9. TF-IDF and cosine similarity from first principles

Problem and idea

Raw sentences cannot be compared numerically. TF-IDF builds one coordinate per term or n-gram. A document receives a high coordinate when it uses that feature often but the feature is uncommon across the documents being compared.

For term t and document d , term frequency $TF(t, d)$ counts or transforms occurrences. Document frequency $df(t)$ counts documents containing t . With N documents, a basic inverse-document frequency is:

```

IDF(t) = log(N / df(t))
TFIDF(t, d) = TF(t, d) × IDF(t).

```

The implementation uses scikit-learn's smoothed conventions, but the intuition is unchanged: ubiquitous words receive little identifying weight.

Tiny text example

Documents are maintenance logiciel finance, maintenance logiciel RH, and construction route. Here maintenance and logiciel appear in 2/3 documents, while finance, RH, construction, and route appear in 1/3. The rarer business terms get larger IDF weights. The first two vectors share maintenance/software coordinates; the road document points elsewhere.

For successor linkage, the vectoriser is fitted **inside each anchor's buyer block**. Word features are 1-2 grams. Character features are within-word-boundary 3-5 grams. The pair score is the maximum of word and character cosine. Character fragments help spelling variants, abbreviations, morphology and formatting, but can also make standard framework boilerplate look deceptively similar.

For the technology classifier, the design differs: word unigrams only were selected within grouped folds; character features are not deployed. Do not conflate the two TF-IDF uses.

Cosine mechanism and hand calculation

For vectors x and y :

$$\cos(x, y) = (x \cdot y) / (||x|| ||y||).$$

The dot product sums shared weighted features. Each norm measures total vector length. Division removes document-length scale, so the score depends mainly on direction.

With $x=(1, 1, 0)$ and $y=(1, 0.8, 0)$, dot product is $1 \times 1 + 1 \times 0.8 = 1.8$; norms are $\sqrt{2} = 1.4142$ and $\sqrt{1.64} = 1.2806$. Cosine is $1.8 / (1.4142 \times 1.2806) = 0.9939$. The texts have almost the same feature pattern. That is strong similarity evidence, not proof of succession: two framework notices can share language without continuing the same need.

10. The four active linkage methods

All methods see the same exposed pairs so differences reflect decision rules, not easier candidate pools.

Method	Mechanism	Acceptance	Strength	Weakness	Locked result
M_A deterministic	verified SIREN AND CPV continuity >0 AND text ≥ 0.25 ; rank by weighted score	fixed gates	auditable strong evidence	misses name-only true links; still accepts generic similar work	15 accepted, precision .533, recall .444
M_B text ranking	rank by max word/character TF-IDF cosine	top score ≥ 0.70	simple, conservative, no unreliable duration	boilerplate/template risk; ignores CPV in decision	8 accepted, precision .875, recall .389
M_C weighted-gated	buyer .50 + text .25 + CPV .20 + time .05, renormalised over observed components; independent gates	weighted score ≥ 70 plus buyer/text/CPV rules	combines evidence, higher recall	hand-chosen weights and more false events	23 accepted, precision .522, recall .667
M_D Fellegi-Sunter	discretise buyer/-text/CPV/time; estimate match and non-match level probabilities by EM; sum log likelihood ratios	posterior $\geq .65$, rank by posterior	data-estimated evidence weights, missing is its own level	conditional-independence and calibration problems	5 accepted, precision .200, recall .056

For M_C, if observed components are set o , its score is:

$$100 \times \sum(k \text{ in } o) w_k c_k / \sum(k \text{ in } o) w_k.$$

If buyer=1, text=.8, CPV=.5 and time is missing, the score is:

$$100 \times (.50 \times 1 + .25 \times .8 + .20 \times .5) / (.50 + .25 + .20) = 84.21$$

Missing time is not treated as time disagreement.

Fellegi-Sunter estimates $m_{kl}=P(\text{level } l \mid \text{match})$ and $u_{kl}=P(\text{level } l \mid \text{non-match})$. A level contributes $\log_2(m_{kl}/u_{kl})$. Evidence common among matches but rare among non-matches is positive; equally common evidence contributes zero. The model was retained as a comparator and did not outperform M_B.

11. Primary top-rank rule, abstention and threshold choice

For each anchor:

```
j_hat_i = argmax over j in J_i of T_ij
Y_i = 1 if T_i, j_hat_i ≥ 0.70, else 0.
```

argmax means “choose the candidate with the largest score.” Example: B=.43, C=.68, D=.76 → D ranks first and is accepted. If B=.41 and C=.66 → C ranks first but is rejected, so the algorithm abstains.

A threshold is needed because every nonempty candidate set has a maximum even when all candidates are bad. Raising the threshold usually decreases accepted volume and recall while increasing precision; lowering it does the reverse. The project retains:

Rule	Events / 3,800	Event rate	Median among events
M_B @ .80	296	7.79%	35.71 months
M_B @ .70	544	14.32%	31.82 months
M_B @ .60	853	22.45%	26.55 months
M_C @ .70	1,332	35.05%	26.09 months

The .70 point is a frozen post-development operating convention, not mathematical truth or an optimiser output. Project history shows that the regional reference informed the retained policy, so the recorded locked stratum is internal validation rather than an untouched holdout. Replacing the policy requires fresh independent evidence.

Why precision-first is defensible

A false positive invents an event **and a date**, producing an unjustified downward step in survival. A false negative leaves a true event censored, pushing survival upward. Both matter. Precision-first is defensible because fabricated exact event times are especially damaging, but it costs recall and observed event count. Because the biases oppose each other, the measured probability is not automatically a lower bound.

12. Reference sample, confusion matrix and uncertainty

The regional reference reviewed 120 anchors on 11 August 2026 before current linkage methods existed. Of these, 112 re-resolve uniquely, 88 are usable, 16 form a pilot split (5 reviewed successors), and 72 form the locked split (18 reviewed successors). The pair tables contain 5,221 pilot and 20,917 locked rows; only 69 locked anchors have an exposed pair, while anchor metrics correctly retain all 72.

Labels came from one LLM research pass over BOAMP notices, official URLs and wider public sources, then subset spot-checking by the owner. They are method-independent reference labels, not legal ground truth or independent specialist validation.

Exact-successor confusion accounting

- TP: accepted candidate equals the reviewed successor.
- FP: an accepted candidate is wrong. A wrong choice on a positive anchor is an FP.
- FN: a reviewed successor is not correctly recovered. A wrong choice on a positive anchor is also an FN.
- TN: a reviewed negative anchor receives no accepted link.

For M_B @ .70: TP=7, accepted=8, so FP=1; 18 positives imply FN=11; all 54 negative anchors are abstained, so TN=54 and negative-anchor FPR=0/54. The materialised JSON separately names **10 abstained positive anchors** and **1 wrong-successor positive anchor**. One narrative report prints FN=10 while also printing recall 7/18; the exact-successor FN is 11. This guide uses the arithmetic and the JSON field definitions rather than silently reconciling the discrepancy.

Metric calculations

Precision = $TP / (TP + FP) = 7 / 8 = 0.875 = 87.5\%$.

Recall = $TP / (TP + FN) = 7 / 18 = 0.3889 = 38.9\%$.

FPR on negative anchors = $FP_negative / (FP_negative + TN) = 0 / (0 + 54) = 0\%$.

Coverage = $accepted / anchors = 8 / 72 = 11.1\%$.

Precision asks whether accepted links are usually correct. Recall asks whether reviewed successors are recovered. A cautious system can have high precision and low recall. Zero observed FPR on 54 negatives is not population-zero FPR; the negatives are corpus-relative and the sample is small.

Wilson confidence intervals

The project uses Wilson intervals for locked precision and recall. With proportion $p_hat = x/n$ and 95% normal quantile $z = 1.96$:

centre = $(p_hat + z^2 / (2n)) / (1 + z^2 / n)$

margin = $z \sqrt{(p_hat(1-p_hat)/n + z^2 / (4n^2)) / (1 + z^2 / n)}$.

For 7/8, the result is **0.529-0.978**; for 7/18, **0.203-0.614**. Wilson pulls the centre away from an unstable boundary and remains inside $[0,1]$. The wide precision interval reflects only eight accepted decisions. A 95% confidence procedure covers the fixed population parameter in 95% of repeated comparable samples; it does not mean a 95% probability that this one fixed interval contains the truth.

The separate 20-link model-assisted production diagnostic found 14 yes, 5 no and 1 uncertain: conservative precision **14/20=.700**, exact Clopper-Pearson CI **.457-.881**, below the .80 target. It is diagnostic, not independent human validation.

13. Turning links into right-censored survival observations

Problem, idea and method

An ordinary average of the 544 linked delays drops 3,256 episodes with no accepted event and therefore conditions on experiencing the event. Survival analysis preserves what those episodes reveal: each remained event-free for at least its observed follow-up.

For an accepted successor:

$\tau_{i,j} = v_{j_hat} - u_i; Y_i = 1.$

For no accepted successor by cutoff $C=2025-12-31$:

$\tau_{i,j} = C - u_i; Y_i = 0.$

$Y=0$ means right-censored, not “never re-procured.” We know only that no accepted BOAMP successor was observed by the cutoff.

Three-row hand example

Episode	Award	Accepted successor/cutoff	Months	Event
A	Jan 2020	successor Jan 2022	24	1
B	Jan 2021	no link by Dec 2025	59	0
C	Jan 2024	successor Jul 2025	18	1

The event-only average is $(24+18) / 2 = 21$ months and says nothing about B's 59 event-free months. Survival methods use all three.

Actual output is one row for each of **3,800** cohort episodes: **544 events** and **3,256 censored**. No duplicate episode and no negative duration exists.

14. Kaplan-Meier and conditional future probabilities

Problem and estimator

The question is: what proportion remains without an accepted observable successor as procurement age grows? Define $S(t) = P(T > t)$. At each event time t_k , let n_k be episodes at risk just before it and d_k events at that time. Kaplan-Meier multiplies conditional survival steps:

$$\hat{S}(t) = \text{product over } t_k \leq t \text{ of } (1 - d_k/n_k).$$

Five-episode example

Suppose times/status are (2, event), (3, censored), (4, event), (5, event), (6, censored) months.

- At month 2, $n=5$, $d=1$: survival becomes $1 \times (1-1/5) = 0.8$.
- Month 3 censoring creates no step; it reduces the later risk set.
- At month 4, three remain at risk, $d=1$: $0.8 \times (1-1/3) = 0.5333$.
- At month 5, two remain, $d=1$: $0.5333 \times (1-1/2) = 0.2667$.

Censoring removes future exposure after its censor time without pretending an event occurred.

Actual results

Age	No-successor survival $\hat{S}(t)$	Cumulative successor probability $1 - \hat{S}(t)$
12 months	.953791	4.621%
24 months	.932669	6.733%
36 months	.913163	8.684%
48 months	.845003	15.500%
60 months	.824647	17.535%

The Kaplan-Meier median is the first time $\hat{S}(t) \leq .5$; it is **not reached**. The reported **31.82 months** is the median delay among the 544 accepted events only. It can exist even while well over half the full cohort remains event-free/censored.

Conditional probability

For an episode event-free at age t , the probability of an event within the next h is:

$$P(T \leq t+h \mid T > t) = 1 - \hat{S}(t+h) / \hat{S}(t).$$

Example at age 36 for the next 12 months: $1 - .845003 / .913163 = .07464 = 7.464\%$. This is different from the unconditional 48-month cumulative probability of 15.500% because it conditions on having reached month 36 without an event.

Actual age-36 estimates are **7.464%** in the next 12 months (95% episode-bootstrap CI **6.522-8.587%**) and **9.693%** in the next 24 (**8.579-11.117%**), based on 500 bootstrap draws. They estimate an observable successor under the primary linkage rule, not a named contract's legal renewal probability.

15. Cox proportional hazards: association, not probability or cause

Problem and intuition

Kaplan-Meier describes groups but cannot simultaneously compare CPV, region, framework status, identifier availability and award year. The hazard is the instantaneous event tendency at age t among episodes still event-free. Cox specifies:

$$h(t|X) = h_0(t) \exp(X' \beta); \text{ hazard ratio } HR = \exp(\beta).$$

$h_0(t)$ is an unspecified baseline shape; X contains covariates; β contains log-hazard coefficients. Exponentiation turns an additive coefficient into a multiplicative hazard ratio.

If $HR = 1.55$, then at the same procurement age and conditional on model covariates, episodes still at risk have an estimated event hazard about 55% higher than the reference. This does **not** mean 55% greater renewal probability, 55% more contracts, 55% shorter duration, or a causal effect.

Reference categories and actual estimates

CPV-32, Bretagne, non-framework, no validated SIREN and centred award year form the relevant baselines. Main results on 3,800 episodes/544 events:

Covariate	HR	95% CI	p-value	Reading
Framework agreement	1.751	1.435-2.136	3.38e-8	higher observed-successor hazard; partly detectability
Validated SIREN	1.082	.885-1.323	.443	no clear association
Award year +1	1.107	1.066-1.149	1.21e-7	time/cohort effect; PH violated
CPV-35 vs 32	1.553	1.218-1.981	.000383	most robust comparative result
CPV-48 vs 32	.828	.638-1.073	.153	not statistically clear
CPV-72 vs 32	1.056	.850-1.310	.624	not statistically clear
Normandie vs Bretagne	.800	.640-1.000	.0501	borderline
Pays de la Loire vs Bretagne	1.003	.815-1.234	.979	no difference

A p-value tests compatibility with $HR=1$ under the model; the confidence interval shows effect uncertainty. Statistical significance is not practical importance or causality.

Proportional-hazards tests flag award year, framework and validated SIREN ($p < .05$). Their reported HRs are time-averaged descriptive associations. The model's in-sample C-index is .626, but out-of-time discrimination is the more honest predictive test.

16. Detectability, C-index and why Cox is not a forecasting model

Candidate-pool detectability

An anchor with 500 candidates has 500 chances to generate a large maximum score; one with three has only three. Thus pool size can affect whether an event is **detectable** even with identical underlying re-procurement behaviour.

The standardised mean difference is:

$$SMD = (\text{mean}_{\text{linked}} - \text{mean}_{\text{censored}}) / \text{pooled_SD}.$$

It measures difference in standard-deviation units, not statistical significance. For log candidate-pool size, linked mean=4.787, censored mean=3.972 and $SMD=+0.470$, the largest measured imbalance.

Adding $\log(1+\text{pool size})$ to a sensitivity Cox model gives HR **1.184** ($p=6.0e-9$). CPV-35 moves only **1.553→1.512**; framework moves **1.751→1.617**, about 14% attenuation on the log-HR scale. This suggests the CPV-35 comparison is largely insensitive to detectability, while part of the framework association reflects buyers with more publication opportunities. The covariate is not causal, so this sensitivity model does not automatically replace the pre-specified primary model.

C-index

For comparable pairs, the C-index asks whether the episode assigned higher risk experiences the event sooner. A predicted-high A with event at 20 months versus predicted-low B at 50 is concordant; reversing the predictions is discordant. $C=.5$ is approximately random ranking and $C=1$ perfect.

Train 2015-2021: 2,470 episodes/392 events, $C=.606$. Test 2022-2024: 1,004/107, **$C=.479$** . The sensitivity test through 2025 is 1,330/152, **$C=.518$** . Therefore the covariates may describe population-level associations while failing to rank individuals on unseen years. A reliable forecast would need stronger external temporal performance, calibration at useful horizons, model updating, richer predictors and independent outcome validation.

17. Parametric survival and robustness of the constructed event

Parametric families

Parametric models assume a distributional family for event time. The project fits exponential (constant hazard), Weibull (monotone hazard), log-logistic, log-normal and generalized gamma (flexible shapes). Maximum likelihood finds parameters that make observed events and censoring most plausible.

$$\text{AIC} = 2k - 2 \log L$$

$$\text{BIC} = k \log n - 2 \log L.$$

k is parameter count and L maximised likelihood. Smaller balances fit against complexity. Results: generalized gamma AIC **7520.99**, then log-normal **7573.24**, log-logistic **7618.57**, Weibull **7633.35**, exponential **7661.31**. Best AIC/BIC does not make generalized gamma literally true. KM supplies all reported within-window operational probabilities; generalized gamma is only the preferred instrument for any cautious extrapolation beyond observation.

Event-definition sensitivity

Different linkage rules create different event sets, hence different survival data. Under the four arms, 12-month successor probabilities are 2.368%, 4.621%, 8.005% and 12.214%; 24-month probabilities are 3.235%, 6.733%, 11.470% and 17.984%. Absolute levels are therefore highly linkage-sensitive.

Two additional checks isolate different mechanisms:

- **Borderline exclusion:** remove 280 anchors whose best score is .65-.75 (133 events, 147 censored). Remaining 3,520/411 give KM 12m **3.721%**, CPV-35 HR **1.780**, framework HR **1.616**.
- **Template/shared-successor re-censoring:** re-censor 173/544 accepted links with word similarity <.50 or a successor reused by another anchor. Events fall to 371 and KM 12m **2.639%**, but CPV-35 HR **1.541** and framework HR **1.692**.

These checks support comparative directions more strongly than absolute levels. They do not prove flagged links false; they bound sensitivity to known signatures.

18. Calendar-time trends: OLS, PELT, stationarity and HMM

Quarterly counts and OLS

Each awarded episode contributes once to its award quarter. Episode counts avoid changes driven merely by corrections or repeated notices. The CPV series uses 43 complete quarters, 2015Q2-2025Q4; partial 2015Q1 is excluded.

For the latest 12 quarters:

$$Y_t = \alpha + \beta t + \text{error}_t.$$

β is episodes per quarter per quarter. Overall $\beta = -.108$ ($p = .921$); CPV-48 $\beta = -.836$ ($p = .0317$); other raw p -values exceed .10. This is a local recent-direction question, not full-history growth or a forecast.

PELT

A straight line can hide level shifts. PELT minimises within-segment squared error plus $\text{penalty} \times \text{number of changes}$. Small penalty over-segments; large penalty suppresses breaks. The central penalty is $\log(n)$ after z-standardisation, with .5 $\times$ and 2 $\times$ sensitivity. A break is stable if all penalties place one within a quarter. Stable candidates are CPV-32 2020Q2, CPV-48 2024Q1 and CPV-72 2021Q1. PELT locates statistical shifts, not their economic causes.

ADF and KPSS

ADF null: unit root/non-stationarity. KPSS null: level stationarity.

ADF	KPSS	Reading
reject	do not reject	supports stationarity
do not reject	reject	supports non-stationarity
reject	reject	conflicting evidence
do not reject	do not reject	inconclusive

Overall rejects ADF ($p \approx 2.9 \times 10^{-7}$) and does not reject KPSS at reported bound .10, supporting stationarity. CPV-32 and CPV-72 reject both, so evidence conflicts. CPV-48 fails to reject ADF ($p = .406$) and rejects KPSS ($p = .0316$), supporting non-stationarity. Short, noisy series make disagreement unsurprising.

HMM

A three-state Gaussian hidden Markov model is fit to quarter-over-quarter count changes. Hidden state is decline/plateau/growth; the observed emission is the numerical change; transition probabilities express persistence/movement; posterior probability quantifies current-state uncertainty. Current regimes: overall growth probability .750, CPV-32 growth .992, CPV-72 growth .594.

HMM, PELT and OLS can disagree without contradiction: HMM describes the most recent latent change regime, PELT partitions historical levels, and OLS averages a 12-quarter slope.

19. Multiple testing and the actual trend verdict

Testing five segments at 5% creates multiple opportunities for a chance small p-value. Under 20 independent nulls, probability of at least one false positive is $1 - .95^{20} = 64.2\%$.

Holm controls family-wise error: sort p-values; compare the smallest with α/m , then progressively less strict thresholds. Adjusted Holm p-values can be computed as running maxima of $(m - \text{rank} + 1) \times p_{\text{sorted}}$, capped at 1. Benjamini-Hochberg controls expected false-discovery proportion: sort p-values, compare $p_{(k)}$ with $k \alpha/m$, and enforce monotone adjusted values.

For five CPV trend slopes, CPV-48 raw $p = .0317$ becomes Holm and BH **.1585**. It is a nominal monitoring signal, not a multiplicity-robust finding. No CPV segment survives correction.

20. Why technology classification is a separate enrichment problem

CPV divisions are broad administrative containers. Cybersecurity, cloud, ERP/business software, telecommunications, AI, data/BI and IT services can share CPV divisions. The classifier predicts a finer business taxonomy from object text; it does **not** create or validate successor links.

The labelled corpus contains 500 unique notices, 2015-2025, in 11 classes: eight substantive technologies plus MIXED, OTHER_DIGITAL and OTHER. Supports are Cloud 32, Cybersecurity 54, Network/Telecom 81, IT Infrastructure 46, Business Software 88, Data/BI 29, AI 7, IT Services 74, Mixed 21, Other Digital 53 and Other 15. Class counts are annotation quotas, not market prevalence. The file has no annotator identifier or second pass, so no inter-annotator agreement exists.

Leakage and grouped folds

Random notice splitting could train on "Maintenance plateforme RH 2022" and validate on its near-identical rectification. Families are unions of shared canonical episode and object character cosine $\geq .80$. This produces 459 groups: 423 singleton, 36 multi-notice, largest 3; 39 near-duplicate pairs are merged, 29 across different reconstructed episodes. Every family stays in one of three stratified folds. Grouping can sacrifice independent signal by merging two true procurements, but that is safer than leakage-inflated accuracy.

21. Multiclass logistic regression, class weighting and macro-F1

Mechanism

Word-TF-IDF converts each object into vector x . Each class k has weights β_k ; score $\beta_k'x$ becomes a probability through softmax:

$$P(Y=k|x) = \exp(\beta_k'x) / \sum_l \exp(\beta_l'x).$$

The largest probability supplies the predicted class. Positive feature weights raise a class score; negative weights lower it. L2-style regularisation constrains large coefficients in the high-dimensional sparse vocabulary. $C=3$ controls inverse regularisation strength in the deployed model.

Class weighting is not feature scaling. Scaling changes numeric feature magnitudes; class weighting gives mistakes on rare classes more loss influence; resampling changes the training rows. With Cloud 150 and AI 7 in a fictional corpus, an unweighted optimiser can improve total loss mostly by serving Cloud. Balanced weights counter that dominance without duplicating AI rows.

For class k , $F1_k = 2 \cdot P_k \cdot R_k / (P_k + R_k)$. Macro-F1 is the unweighted average over 11 classes. A model predicting a 90%-majority class can score 90% accuracy but nearly zero F1 on minorities; macro-F1 exposes that failure.

Model comparison

Model	Representation	Mean grouped-CV macro-F1	Fold SD
Majority	constant	.027	.0004
CPV	administrative	.439	.076
CPV+descriptor	administrative	.469	.038
logistic	word TF-IDF	.662	.067
balanced logistic	word TF-IDF	.741	.034
linear SVM	word TF-IDF	.717	.030
balanced SVM	word TF-IDF	.716	.054

Balanced logistic is selected because it leads mean macro-F1, has the smallest train-validation gap among text models (.249), stable representation choices, good temporal behaviour and native probability output. Every fold chose word unigrams rather than the offered 1-2 gram representation. CamemBERT was not run because the pre-specified gate required classical macro-F1 below .55; observed .741 exceeded it.

CPV is deliberately excluded from text features so the benchmark asks a scientific question: can text add a business taxonomy beyond official administrative coding?

22. Bootstrap, calibration, confidence and technology quality gates

Family bootstrap and paired comparison

To approximate sampling variation, sample 459 procurement families with replacement, rebuild the evaluation rows, calculate macro-F1, and repeat 1,000 times. Families, not notices, are resampled because related notices are not independent.

Selected model macro-F1 is **.7442**, bootstrap SD **.0275**, 95% percentile CI **.6819-.7905**. CPV/descriptor is **.4731**, CI **.4126-.5259**. In each replicate compute $D_b = F1_{\text{text},b} - F1_{\text{CPV},b}$; observed difference **.2711**, CI **.2009-.3403**, excluding zero. Pairing isolates the within-sample performance difference better than comparing two independent intervals.

Calibration decision

Accuracy and calibration differ. A model can rank correctly while its .70 score does not correspond to 70% empirical correctness. Platt/sigmoid calibration was evaluated inside grouped splits. Pre-specified adoption required ECE improvement $\geq .02$ and macro-F1 cost $\leq .02$. ECE improved by .1405, but macro-F1 fell by .0364, so calibration was rejected. The deployed score is raw, uncalibrated class probability.

At raw score $\geq .70$, out-of-fold accuracy is **43/45=95.56%** on the annotation distribution, versus 74.95% below. In production, **235/3,800=6.18%** episodes clear .70. The score is useful for ordering/filtering but is not automatically the probability a deployed prediction is correct because it is miscalibrated and class priors differ between quota corpus and population. The technology .70 and linkage .70 are numerically coincident but solve unrelated problems.

Gates

```
Predicted class
  ↓ substantive technology, labelled support ≥10, OOF F1 ≥.65?
Classifier Gate A
  ↓ survival: ≥100 episodes and ≥20 events?
Statistical Gate B
  ↓
Include in technology survival / exclude with reason
```

Five classes pass both: Cybersecurity (316 episodes/60 events), Network/Telecom (859/140), IT Infrastructure (298/38), Business Software (854/106), IT Services (492/72). Cloud and Data/BI fail downstream

sample/event support; AI fails labelled support; residual classes are not substantive technologies.

23. Technology survival and trend results

The predicted episode class becomes a constructed covariate. Technology-specific KM curves compare time to the already-defined observable successor. The log-rank null is that all included survival functions are equal over time.

Among five gated classes (2,819 episodes, 416 events), log-rank statistic **10.258**, $p=.0363$. At 24 months, Business Software has 8.57% estimated successor probability and IT Infrastructure 6.11%. This omnibus result says not all five curves are alike; it does not identify a causal technology effect or a specific pair, and it is unadjusted for buyer/procedure/size. Including statistically large but substantively residual classes would produce $p=.000119$; the gate prevents that heterogeneous fallback contrast from masquerading as a stronger technology finding.

Technology trends use all predictions because the $\geq .70$ subset leaves mostly empty quarterly series. They use the same 43-quarter observation window as the CPV series (2015Q2-2025Q4), with the reported OLS slopes fitted over the latest 12 quarters. Five gated classes are tested: the smallest raw p is IT Services **.1944** (slope **+.4476** episodes per quarter; Holm **.9720**), so no class is even nominally significant. This is a negative result: there is no corrected evidence of a recent linear technology-count trend. Classification uncertainty is not propagated into trend or survival intervals, so both branches inherit additional measurement error.

24. What could threaten the analysis?

Threat	Evidence / calculation	Why it matters	Decision	Residual limitation
Missing duration	2,847/3,800=74.9%; 2023 11.8% vs 2025 84.4% complete	expiry/timing rule would be schema-driven	no global imputation; removed duration-conditioned arm	cannot study true contract duration broadly
Missing validated identity	2,521/3,800=66.3%	name-only buyer errors	keep names but veto conflicting validated SIRENs	legal-form changes can be missed
Post-award information / unequal follow-up	recent cohorts have short follow-up; 2025 Q4 cannot show ≥ 90 -day event	censoring and award year intertwine	administrative censoring; 2025 excluded from primary temporal test	independent censoring may be imperfect
Text-length regime	linked mean 1,611 chars vs censored 1,087; SMD .262	more text improves matchability	report detectability diagnostics	content and measurement cannot be separated fully
Multi-lot episodes	1,176/3,800 enter on digital lot; 412 multi-digital	broad episode mixes domains	any-code rule + explicit segment tie-break	segment can simplify mixed demand
Unresolved award date	26/3,826 awarded digital episodes dropped	no defensible time zero	drop rather than guess	selection toward better records
Candidate-pool imbalance	log pool SMD .470	maximum-over-more-draws bias	pool-adjusted Cox sensitivity	residual buyer publication behaviour
Shared successors	127 links share successor; 173 total template-risk flags	no one-to-one constraint may reuse generic notice	re-censor sensitivity	flagged $\neq$ false; exact rate unknown
Boilerplate similarity	65 links have word score $<.50$ but character score carries acceptance	framework text can look alike	template-risk re-censoring	semantic false positives remain possible
Rare technology classes	AI $n=7$; no interpretable performance	unstable per-class metrics	fail classifier gate	AI conclusions unavailable
Small linkage validation denominator	precision 7/8, CI .529-.978	point estimate unstable	report Wilson interval and production diagnostic	independent specialist accuracy absent
Annotation inconsistency	3 near-duplicate families have conflicting labels	ceiling on classification	leave labels unchanged and document	no inter-annotator reliability

Threat	Evidence / calculation	Why it matters	Decision	Residual limitation
Amount ambiguity	15.7% no container; multiple unvalidated candidates	false monetary trends	omit monetary analysis	spend/value question unanswered

An upstream error propagates:

wrong episode → wrong anchor → wrong block → wrong link → wrong event time → distorted KM/Cox
 blocking miss → model cannot select → censored row → event level pushed down
 false accepted link → invented event/date → event level pushed up

Because the last two mechanisms point in opposite directions, observed successor probabilities are not guaranteed lower bounds.

25. Metric and calculation dictionary

Quantity	Type	Formula/algorithm	Actual value and denominator	Interpretation / limitation
Standardised notices	observed→standardised	unique official ID	1,620,712/1,620,712 unique	publications, not contracts
Episodes	constructed	graph connected components	1,103,632 from 1,620,712 notices	inferred processes
Cohort	constructed	region+award+any digital CPV+date	3,800; 26 unresolved-date drops	selected study population
Digital-lot/main-outside share	constructed	1,176/3,800	30.95%	why “any code” matters
Candidate coverage	constructed	3,520/3,800	92.63%	any candidate, not correct candidate
Candidate pairs	constructed	anchor-candidate rows	763,417; median 89 per covered anchor	exposed comparisons
Blocking ceiling	estimated on reference	21/23	91.30%	not fully surfacing-independent
Primary events	constructed	M_B top score $\geq .70$	544/3,800	accepted observable successors
Event rate	descriptive	544/3,800	14.3158%	linkage-conditioned
Observed-event median	descriptive	median of 544 event delays	31.82 months	not KM median
Precision	estimated on reference	7/8	87.5%, CI .529-.978	accepted-link correctness in locked sample
Recall	estimated on reference	7/18	38.89%, CI .203-.614	recovered reviewed successors
Negative-anchor FPR	estimated on reference	0/54	0%; small corpus-relative denominator	not population zero
KM 12m	estimated	1-product risk steps	4.621%, 3,800/544	event probability under primary linkage
KM 24m	estimated	same	6.733%	linkage-sensitive
KM median	estimated	first $S \leq .5$	not reached	differs from event-only median
CPV-35 HR	estimated/associational	exp Cox coefficient	1.553, CI 1.218-1.981	vs CPV-32, non-causal
Framework HR	estimated/associational	exp Cox coefficient	1.751, CI 1.435-2.136	partly detectability
Pool SMD	descriptive diagnostic	mean difference/pooled SD	+ .470 log pool	detectability imbalance, not p-value
Pool-adjusted HRs	robustness	Cox + log pool	CPV-35 1.512; framework 1.617	sensitivity, not new primary
Out-of-time C-index	predictive	concordant comparable pairs	.479 on 1,004/107	no useful individual ranking
Parametric winner	estimated	minimum AIC/BIC	generalized gamma AIC 7520.99	preferred fit, not truth

Quantity	Type	Formula/algorithm	Actual value and denominator	Interpretation / limitation
Trend CPV-48	descriptive/inferential	12-quarter OLS	slope -.836; raw p .0317; Holm .1585	nominal only
Labelled notices/families	observed/constructed	annotations/grouping	500/459	quota corpus
Technology macro-F1	predictive estimate	mean class F1	.7442; 1,000-family bootstrap CI .6819-.7905	corpus performance
CPV benchmark	predictive estimate	same folds/metric	.4731; CI .4126-.5259	administrative baseline
Paired gain	predictive comparison	F1_text–F1_CPV	.2711; CI .2009-.3403	text adds signal
High-confidence predictions	predicted	raw score $\geq .70$	235/3,800=6.18%	operational subset, not observed truth
Technology survival	associational enrichment	gated log-rank	5 classes; 2,819/416; p .0363	predicted groups, unadjusted
Technology trend	robustness/negative	5 slopes + Holm/BH	none survives; min Holm .2815	no supported linear trend

Complex numbers are not reducible to one fraction. KM iterates risk sets; Cox maximises partial likelihood; C-index enumerates comparable pairs; bootstrap resamples families; AIC/BIC use maximised likelihood; PELT optimises penalised partitions; ADF/KPSS fit time-series regressions with opposite nulls; HMM estimates transition/emission parameters and posteriors; Holm/BH reorder a family of p-values.

26. Why each method was chosen, and what happens if it is removed

Problem	Alternative	Chosen solution	Why / protection	Trade-off; without it
many notices per process	notice-level analysis	episode reconstruction	avoids administrative duplicates	inferred merges; without it false short events
huge pair space	exhaustive comparison	buyer/time blocking	computation and false-opportunity control	ceiling < 1 ; without it noisy billions
no-match anchors	always choose top	threshold + abstention	avoids forced false events	lower recall; without it every covered anchor “renews”
text representation	raw string rules / embeddings	TF-IDF word+character for linkage	transparent, local vocabulary, variants	boilerplate risk; embeddings were not active/evaluated
event-time censoring	event-only mean	Kaplan-Meier	uses incomplete follow-up	independent-censoring assumption
adjusted association	separate group means	Cox	simultaneous covariates, unspecified baseline	PH assumptions; not prediction/causality
tail model	KM extrapolation	generalized-gamma comparison	explicit tail family	extrapolation uncertainty
related annotation rows	random CV	family-grouped CV	blocks near-copy leakage	fewer independent units
class imbalance	accuracy/unweighted loss	balanced logistic + macro-F1	minority classes matter equally	probability calibration distorted
small-sample uncertainty	normal approximation	family bootstrap; Wilson for linkage	respects clustering/bounds	still depends on sample representativeness
many trend tests	raw p-values	Holm and BH	false-positive control	less power
weak predicted groups	analyse all labels	classifier/statistical gates	prevents residual buckets driving conclusions	excludes potentially interesting small classes

Some rationales are explicit project decisions: frozen precision-first threshold, any-code cohort, group-aware CV, no duration imputation and gates. Standard explanations of why KM, Cox, bootstrap, Holm and BH behave as described are statistical theory used to teach the documented choice, not invented project history.

27. Assumptions and consequences if they fail

Assumption	Where	Need	Failure consequence	Tested?
episode edges join one process	reconstruction	define analysis unit	duplicated/merged needs	structural tests; semantic spot-check incomplete
buyer/time block contains true links	candidates	feasible search	unrecoverable false negatives	21/23 reference reachability
high text similarity tracks same need	M_B	select successor	boilerplate false links	reference + template-risk sensitivity
administrative censoring is non-informative conditional on design	KM/Cox	use incomplete follow-up	biased curves if linkability/follow-up differs	diagnostics, not fully resolved
proportional hazards	Cox	constant HR interpretation	time-averaged/misleading coefficient	violations found for 3 covariates
chosen covariates address relevant confounding for association	Cox	adjusted comparisons	residual confounding	no causal design
parametric family approximates tail	extrapolation	beyond-window estimates	wrong tail probabilities	AIC/BIC/KM fit comparison only
quarterly counts are comparable	trends	interpret change	schema/publication shifts mimic demand	duration break documented; no causal attribution
OLS errors and form support slope inference	trend	p-values	unreliable significance	descriptive only
PELT cost/penalty reflect meaningful level shifts	change points	parsimonious breaks	over/under-detection	three penalties
HMM Gaussian state model is adequate	regimes	latent-state summary	unstable/mislabelled states	posterior reported; limited series
annotations reflect frozen taxonomy	classifier	supervised target	model learns inconsistent boundaries	three conflicting families documented
families are independent resampling units	CV/bootstrap	uncertainty/leakage	intervals too narrow if dependency remains	grouping audit
future text resembles labelled years	deployment	generalisation	temporal drift	2023-25 holdout, limited rare-class support

28. Which conclusions are strongest and weakest?

Stronger / relatively robust

- The software/data pipeline is reproducible and structurally coherent: canonical validation passed; all notices map once; counts reconcile.
- BOAMP notices must be converted to episodes before successor analysis; the 31.9% publication-to-episode reduction demonstrates the grain problem.
- Text contains technology information beyond CPV: paired macro-F1 gain .271 with CI excluding zero on identical grouped folds.
- CPV-35 has a higher observable-successor hazard than CPV-32: HR direction survives threshold arms, borderline removal, template-risk re-censoring and pool adjustment.

Moderate / descriptive

- Under M_B @ .70, 544/3,800 episodes have accepted observable successors and KM 12m is 4.621%. Exact levels are well computed for that event rule but not stable across reasonable linkage rules.
- Five gated technology curves differ omnibus ($p=.0363$), but groups are predicted and the test is unadjusted.
- PELT and HMM provide monitoring signals, not causes or forecasts.

Fragile / sensitivity-dependent

- Framework HR magnitude: direction is stable, but detectability attenuates 1.751→1.617 and PH is violated.
- Absolute successor probability and event rate: 7.79%-35.05% event rate across retained arms.
- Locked precision .875: only eight accepted links and CI .529-.978; separate diagnostic gives .700 conservatively.
- CPV-48 recent decline: raw p .032 but adjusted .159.

Not established

- Confirmed legal renewals, population linkage accuracy, causal effects, individual contract forecasts, market shares from predicted classes, AI classifier performance, awarded-amount trends, and causes of change points.

29. The entire project as one connected story

We begin with BOAMP because it records official procurement publications. But BOAMP does not identify renewals and one procurement can appear through several notices. We first standardise schema-specific fields while keeping raw lineage, then reconstruct notices into episodes so administrative repeats do not become new demand.

We restrict episodes to awarded Grand Ouest procurement containing at least one qualifying digital CPV code and a valid award date. Each selected award becomes an anchor. We search forward 90-2,920 days among plausible same-buyer episodes, vetoing conflicting validated SIRENs. This blocking step creates comparisons but can permanently lose true successors.

For every anchor block, word and character TF-IDF turn descriptions into vectors and cosine similarity measures their alignment. Four linkage methods are evaluated on the same candidates. The frozen primary rule chooses the most text-similar candidate only if its score reaches .70; otherwise it abstains. The output is an observable-successor proxy, never a legal-renewal label.

Accepted links provide event dates. Abstentions provide follow-up until 31 December 2025 and are right-censored. Kaplan-Meier estimates linkage-conditioned survival; Cox describes adjusted associations; parametric families characterise possible tails. Threshold, borderline, template-risk and candidate-pool checks show that comparative directions-especially CPV-35-are more stable than absolute probability levels. Out-of-time C-index shows the Cox model should not be used for individual prediction.

Calendar-quarter aggregation answers a separate question about activity through history. OLS summarises recent direction, PELT flags candidate structural breaks, ADF/KPSS assess stationarity evidence, and an HMM describes current latent change regimes. Multiplicity correction turns CPV-48 into a nominal monitoring signal rather than a finding.

Finally, a labelled text corpus supplies a finer business-technology taxonomy than CPV. Grouped folds prevent related-notice leakage; balanced multinomial logistic regression wins a pre-specified comparison; family bootstrap quantifies uncertainty; calibration is evaluated and rejected by rule. Predictions enrich all episodes, but quality and sample-size gates restrict downstream survival/trend analysis. The final story is scientifically useful because it owns every transformation and limitation: it measures visible re-procurement patterns under a documented observation process, not legal renewal, causality or individual destiny.

30. Four levels of understanding**Level 0 - one sentence**

I built and validated a BOAMP pipeline that reconstructs procurement episodes, links plausible later successor procurements, analyses their censored timing, and enriches them with predicted technology classes-while keeping the results explicitly distinct from legal renewal and causal prediction.

Level 1 - 60 seconds

BOAMP publishes notices, not clean contracts or renewals. I standardised 1.62 million notices and reconstructed 1.10 million procurement episodes, then selected 3,800 awarded Grand Ouest episodes containing a digital CPV. For each award I generated later same-buyer candidates and used a frozen, conservative TF-IDF text-ranking rule to accept at most one observable successor. That produced 544 events; the rest

were right-censored, so I used Kaplan-Meier and Cox rather than averaging only linked cases. The main comparative finding is a higher observed-successor hazard for CPV-35, but absolute probabilities depend strongly on the linkage rule and the Cox model has weak out-of-time ranking. Separately, a grouped-CV technology classifier substantially outperformed CPV as a business taxonomy. The project is descriptive and associational, not legal-renewal proof or causal forecasting.

Level 2 - five minutes

Explain the chain in Section 2, the 3,800 cohort funnel, the 90-2,920-day block, local word/character TF-IDF cosine, top-1 .70 abstention, reference precision 7/8 and recall 7/18 with wide Wilson intervals, 544/3,800 events, right censoring, KM 12m 4.621% and median not reached, CPV-35 HR 1.553 with pool-adjusted 1.512, out-of-time C=.479, four event-definition arms, and the separate 500-notice/459-family technology classifier with macro-F1 .744 vs .473 CPV baseline.

Level 3 - technical defense

Use Sections 3-28: define each analytical grain and symbol; derive TF-IDF/cosine, Wilson, KM, conditional probability, Cox HR, SMD, C-index, AIC/BIC, OLS, PELT objective, ADF/KPSS nulls, HMM state posterior, softmax, macro-F1, bootstrap and multiplicity corrections; then state the exact denominators, assumptions, PH violations, reference provenance, calibration decision, quality gates, contradiction note and robustness results.

31. Oral-defense questions and evidence-backed answers

Challenge	Simple answer	Technical answer	Numerical evidence
Why not notice level?	One procurement publishes several notices.	Notice-level linkage creates self/duplicate and short-gap events; connected components define episodes.	1,620,712 notices → 1,103,632 episodes.
Why episodes?	They approximate one procurement process.	Union-find over folder, explicit links and constrained same-reference edges, with SIREN conflict veto.	517,080 fewer units; zero buyer-conflict episodes.
Why these CPVs?	They define a broad auditable digital scope.	Episode enters if any code division is 32/35/48/72.	7,376 digital-scope Grand Ouest episodes before award filter.
Why any-code inclusion?	Digital work can be one lot of a broader tender.	Main-only would exclude mixed episodes carrying a qualifying lot.	1,176/3,800 (30.9%) have non-digital main CPV.
Why 90 days?	Shorter gaps looked like concurrent work.	Lower floor removes parallel lots; shortest reviewed successor is 139 days.	Earlier pathology: 132/628 links under 3 months.
Why eight years?	It bounds the search, not contract duration.	2,920-day operational horizon excludes no reviewed successor.	reviewed gaps up to 2,644 days.
Why text similarity?	Similar needs often reuse discriminative wording.	Local word 1-2 gram and char 3-5 gram TF-IDF cosine rank same-buyer candidates.	M_B locked precision 7/8.
Why not embeddings?	They were not an active evaluated method.	The project selected transparent sparse features; no canonical comparison supports an embeddings claim.	Four active linkage methods only.
Why .70?	Conservative frozen abstention point.	Set before reference inspection to prioritise false-event control; .60 remains sensitivity.	296/.544/.853 events at .80/.70/.60.
Why precision-first?	False links invent event dates.	FP moves KM at a false time; FN censors a true event.	M_C recall .667 but precision .522 vs M_B .389/.875.
What does 87.5% precision mean?	Seven of eight accepted locked links matched review.	Reference-sample exact-successor precision with Wilson uncertainty.	.875, CI .529-.978, n=8.
Why is recall low?	The rule abstains deliberately and blocking misses some.	TP/18=.389; maximum reference recall 21/23=.913 before scoring.	7 recovered; 10 abstentions + 1 wrong choice.
What is censoring?	Follow-up ends before an event is observed.	Y=0 contributes risk time to cutoff, not a no-renewal label.	3,256/3,800 censored.
Why KM?	It keeps censored episodes.	Product-limit estimator updates at events and removes censoring only from future risk sets.	KM 12m 4.621%; median not reached.
Why Cox?	To compare characteristics jointly.	Semiparametric proportional-hazards model estimates adjusted HRs without specifying baseline hazard.	CPV-35 HR 1.553.
Why isn't HR a probability?	It is a rate ratio among those still at risk.	Probability depends on baseline hazard and time integration as well as HR.	CPV-35 1.553; KM probabilities are separate.
Significant HR but weak C-index?	Group association need not rank individuals.	Coefficients can shift average hazard while overlapping individual risk scores.	CPV-35 p=.00038; test C=.479.
Why not individual prediction?	It fails out-of-time ranking.	Prediction requires discrimination and calibration on unseen cohorts.	2022-24 C=.479; 2022-25 C=.518.
Why bootstrap?	Sampling a different labelled corpus could change F1.	Resample whole procurement families to preserve within-family dependence.	1,000 draws; .744 CI .682-.791.
Why grouped CV?	Near-duplicate notices would leak.	All episode/≥.80-character-similarity family members stay in one fold.	459 families; 29 near-duplicate pairs across episodes.
Why macro-F1?	Rare classes count equally.	Average 11 class-specific harmonic means, not observation-weighted accuracy.	accuracy .766; macro-F1 .744.

Challenge	Simple answer	Technical answer	Numerical evidence
Why not trust classifier probability?	Raw scores are miscalibrated and priors differ.	Calibration improved ECE but violated the macro-F1 loss budget; quota priors do not equal deployment priors.	ECE gain .1405; macro-F1 change -.0364.
Why adjust p-values?	Several chances can create a lucky signal.	Holm controls family-wise error; BH controls FDR.	CPV-48 raw .0317 → .1585.
Why can PELT/HMM/OLS disagree?	They answer different time questions.	historical level partitions vs current latent change vs 12-quarter average slope.	overall OLS $p = .921$ while HMM growth = .750.
Biggest weakness?	The successor event lacks independent specialist validation.	Reference is method-independent but single-pass LLM/subset-checked, with unknown review-candidate surfacing.	precision $n=8$; diagnostic 14/20.
Strongest result?	Text adds technology signal; CPV-35 comparison is robust.	paired grouped-family bootstrap and multi-arm survival sensitivity respectively.	gain .271 CI excludes 0; CPV-35 1.553→1.512.
Most cautious result?	Absolute successor probability and CPV-48 trend.	event rule changes levels; multiplicity removes trend significance.	event rate 7.79-35.05%; .0317→.1585.

32. Recommended next steps and further questions

1. Obtain blinded procurement-specialist review of the frozen accepted-link sample, keeping the audit key hidden until labels are final; do not tune and evaluate on the same rows.
2. Run a compact semantic episode-reconstruction review, prioritising long-span, high-notice-count and reference-conflict episodes.
3. If recalibrating linkage after human review, create a fresh holdout and compare one-to-one/shared-successor controls plus template-resistant text features.
4. Add labelled procurement families, especially AI, Cloud/Data and documented taxonomy boundaries; repeat the same grouped temporal evaluation.
5. Before monetary analysis, define and validate one canonical awarded-value estimand across schemas and lots.
6. Monitor whether CPV-48's nominal decline persists and seek documentary evidence before proposing a cause.

Open questions that could change interpretation are: how much independent review lowers or raises production precision; whether missing successors concentrate among legal-form changes or purchasing-centralisation; whether PH violations reflect genuine time-varying associations; and how classification uncertainty should be propagated into technology survival curves.

Source-of-truth note

This guide is derived from the active canonical repository state, especially:

- PROJECT_WORK_PROTOCOL.md and FINAL_PIPELINE.md;
- DATA_QUALITY_REPORT.md;
- SURVIVAL_ANALYSIS_REPORT.md;
- TREND_ANALYSIS_REPORT.md;
- TECHNOLOGY_TAXONOMY_REPORT.md; and
- the current materialised JSON/CSV summaries under `data/processed/boamp/`.

Archived and retired methods are excluded.